SUEZ Platform – Software Architecture Document

Version 1.0 | Architecture Draft

1. ELI5 — Explain Like I'm 5

Imagine a big company where thousands of employees want to use AI tools like ChatGPT or Claude.

If everyone connects directly:
• The company can't control usage
• Sensitive data may leak
• AI costs become unpredictable
• There is no audit trail

SUEZ acts like a smart security gate in front of all AI systems.

Employee App → SUEZ → AI Provider

SUEZ checks:
• Who is allowed to use AI
• Which models are allowed
• Token limits and policies
• Tool usage permissions
• Cost tracking

SUEZ keeps records so companies know:
• Who used AI
• Which models were used
• How much it cost
• Whether policies were followed

In simple terms: SUEZ is the security guard, accountant, and traffic controller for enterprise AI usage.

2. Purpose of This Document

This document describes the architecture of the SUEZ platform including system components, deployment model, and technical design principles.

Audience:
• Engineering
• Platform Architects
• Security Engineers
• SRE / Infrastructure teams
• Technical leadership

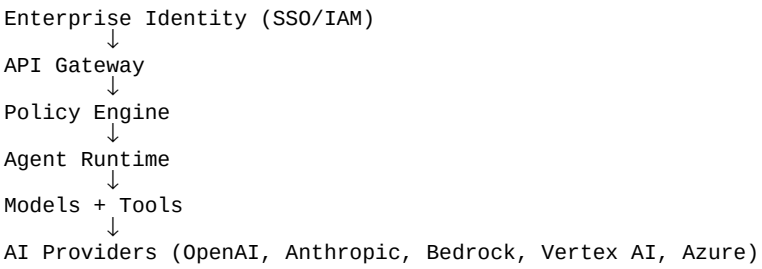
3. System Overview

SUEZ is a multi-tenant enterprise SaaS platform that acts as a governed gateway and control plane for enterprise AI usage.

Responsibilities:
• AI access governance
• Policy enforcement
• Provider abstraction
• Usage metering

- Security and compliance
- Billing and cost visibility
- AI agent orchestration

4. High-Level Architecture



5. Architectural Principles

Provider Abstraction:
Unified interface for multiple AI providers.

Governance by Default:
All requests pass through policy enforcement.

Multi-Tenant Isolation:
Tenant-level data, keys, and policy separation.

Horizontal Scalability:
Designed for 10,000 tenants and 1000 requests/sec.

6. Core Components

API Gateway:
Authentication, rate limiting, routing, TLS termination.

Policy Engine:
Evaluates model permissions, token limits, and provider restrictions.

Resource Management:
Handles lifecycle of Models, Tools, and Agents.

Agent Runtime:
Executes AI workflows combining models and tools.

Usage Metering:
Records tokens, cost, resource usage, and timestamps.

Billing Engine:
Supports subscription and usage-based billing.

7. API Layer

```
Model APIs:
POST /models
GET /models
DELETE /models/{id}

Tool APIs:
POST /tools
GET /tools
DELETE /tools/{id}

Agent APIs:
POST /agents
GET /agents
DELETE /agents/{id}
```

8. Security Architecture

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Identity:
SSO, OIDC, IAM integration.

Data Protection:
TLS everywhere, encrypted secret storage, tenant isolation.

Tool Security:
Scoped credentials, rate limits, audit logging.
```

9. Observability

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Metrics:
Request rate, latency, error rate.

Logs:
Request logs, policy decisions, admin actions.

Tracing:
End-to-end distributed tracing across services.
```

10. Reliability & Performance

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Availability Target: 99.9%
RTO: 1 hour
RPO: 15 minutes

Performance Targets:
Gateway latency overhead <150ms
Policy evaluation <50ms
```

11. Deployment Model

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Cloud-native SaaS architecture typically deployed on Kubernetes.

Components:
• API Gateway / Ingress
• Microservices
```

• Managed databases
• Event streaming
• Observability stack

12. Future Roadmap

Planned Enhancements:
• Agent marketplace
• Tool registry
• Model benchmarking
• AI risk scoring
• Multi-region deployment

13. Summary

SUEZ provides a governed control plane for enterprise AI usage.

The platform enables:
• Secure AI adoption
• Governance and compliance
• Cost visibility
• Multi-provider abstraction
• Scalable enterprise deployment